

Scientists are interested in determining the underlying metabolic reprogramming events that lead to type 2 diabetes mellitus (T2DM), a disease characterized by insulin resistance and increased hepatic glucose production. MicroRNAs (miRNAs) are a class of small noncoding RNAs implicated in T2DM pathology. miR-26a, a member of the miR-26 family of miRNAs, has been shown to play a critical role in regulating hepatic glucose production and lipid metabolism. miR-26a has also been implicated in the prevention of obesity-induced insulin resistance, as evidenced by decreased miR-26a expression in obese, leptin-deficient (*ob/ob*) mice compared to lean control mice.

Experiment 1

Researchers designed an experiment to promote obesity-induced insulin resistance in wild-type (WT) mice and *Alb Tg* mice, which express hepatic miR-26a at levels three-fold greater than WT. WT mice and *Alb Tg* mice were fed either a standard healthy diet (SHD) or a high-fat diet (HFD) for 14 weeks. All mice were then fasted for 13 hours and received an abdominal injection of a glucose preparation (2 g/kg of body weight). The mice were then fasted for 10 hours to allow blood glucose levels to return to baseline, after which an insulin preparation (1 U/kg of body weight) was injected in the abdomen. Blood samples were drawn from the tail veins of all specimens to assess serum glucose levels. Figure 1 shows the experimental results for WT mice and *Alb Tg* mice fed a HFD.

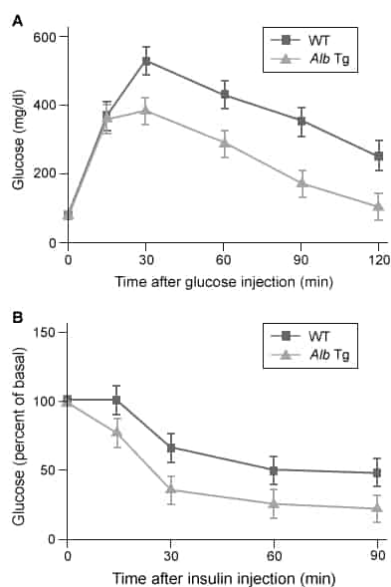


Figure 1 Effect of glucose and insulin administration on blood glucose of WT and *Alb Tg* mice fed a HFD. (Note: WT and *Alb Tg* mice fed a HFD had equal weight gain.)

Experiment 2

Blood samples were taken from fasting human adults, and levels of hepatic miR-26a were measured in comparison to a ubiquitously expressed reference gene, RNU24. Figure 2 shows the miR-26a levels plotted against each subject's body mass index (BMI, the ratio of body mass to squared body height).

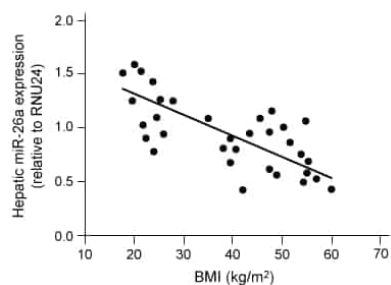


Figure 2 Relationship between BMI and hepatic miR-26a expression

Type 1 diabetes mellitus (T1DM) differs from T2DM in that affected patients stop producing insulin at a young age. Which of the following is most likely impaired in T1DM?

- I. Exocrine function of pancreatic beta cells
- II. Endocrine function of pancreatic beta cells
- III. Exocrine function of pancreatic alpha cells

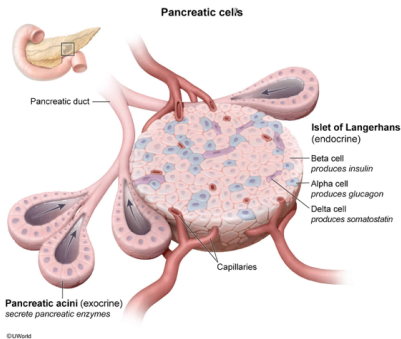
IV. Endocrine function of pancreatic alpha cells

- ☐ A. I only [11%]
☒ B. II only [75%]
☐ C. I and II only [10%]
☐ D. III and IV only [2%]

Correct

75% answered correctly

Explanation:



The pancreas has paracrine, exocrine, and endocrine functions. Cells with **paracrine function** secrete substances that exert

effects on neighboring cells whereas cells with **exocrine function** secrete substances (eg, saliva, sweat, enzymes) through a duct and onto an epithelial surface. In comparison, cells with **endocrine function** secrete hormones into the bloodstream to cause an effect in a different part of the body.

The functional units of the **endocrine pancreas** are the **islets of Langerhans**, which are made up of alpha, beta, and delta cells that secrete different hormones into the blood.

- **Beta cells** produce the hormone **insulin**, which inhibits neighboring alpha cell function; therefore, beta cells also exhibit *paracrine* function.
- **Alpha cells** produce the hormone **glucagon**. Alpha cells also exhibit *paracrine* function, as they are able to inhibit beta cell function in certain settings.
- **Delta cells** produce **somatostatin**, a hormone that has a generalized inhibitory effect on digestive function and has been shown to suppress insulin and glucagon release.

Because the question states that **type 1 diabetes mellitus** (T1DM) is caused by a **failure** in **insulin production**, affected patients most likely have **impaired endocrine** function of pancreatic **beta cells** (**Number II**).

(Numbers I and III) The acinar cells comprise the exocrine pancreas and are responsible for **secreting enzymes** via the pancreatic duct to aid digestion in the duodenum. However, pancreatic alpha and beta cells do *not* display exocrine function.

(Number IV) Failed insulin production in T1DM is due to impaired endocrine function of beta (*not* alpha) cells.

Educational objective:

In the setting of high blood glucose, beta cells release insulin to promote glucose uptake from the blood and inhibit glucagon secretion by alpha cells.

Time spent: 0 sec
Subject:
Biology

QID: 400661
System:
Endocrine and Nervous Systems

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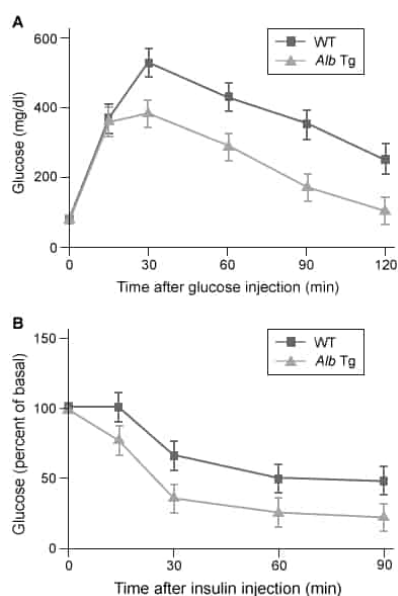


Figure 1 Effect of glucose and insulin administration on blood glucose of WT and *A/b* Tg mice fed a HFD. (Note: WT and *A/b* Tg mice fed a HFD had equal weight gain.)

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Blood samples were taken from fasting human adults, and levels of hepatic miR-26a were measured in comparison to a ubiquitously expressed reference gene, RNU24. Figure 2 shows the miR-26a levels plotted against each subject's body mass index (BMI, the ratio of body mass to squared body height).

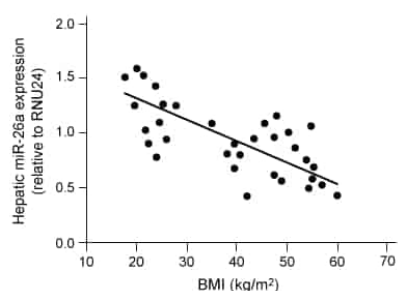


Figure 2 Relationship between BMI and hepatic miR-26a expression

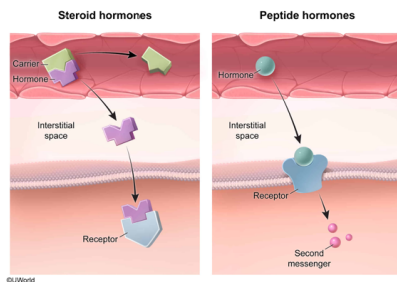
Abdominal injection of insulin would cause the peptide hormone to enter the bloodstream and exert its effect on target cells by:

- ☐ A. acting as a second messenger. [8%]
- ☐ B. diffusing through the plasma membrane. [8%]
- ☒ C. binding a receptor on the plasma membrane. [82%]
- ☐ D. disassociating from a carrier protein. [1%]

Correct

81% answered correctly

Explanation:



The solubility of a hormone in water affects how it is transported in the blood and how it exerts its physiological effect on target cells. Hormones are primarily divided into three different classes, with their chemical structure determining their water solubility:

- **Peptide hormones** (eg, *insulin*, *glucagon*) are composed of amino acids linked by peptide bonds and are made in the rough **endoplasmic reticulum** (ER). Peptide hormones range from small to large depending on their number of amino acids. The existence of an overall charge makes peptide hormones **water-soluble (hydrophilic)**, which means they can dissolve in the bloodstream but cannot cross the **hydrophobic lipid bilayer** of the target cell plasma membrane. Accordingly, peptide hormones **act as first messengers** and must **bind a receptor on the plasma membrane** to induce a signaling cascade via activation of intracellular **second messengers**.
- **Steroid hormones** are lipid hormones derived from cholesterol and generated in the smooth ER. These hormones are fat-soluble (hydrophobic) and must therefore be bound to a carrier protein in the bloodstream. However, their hydrophobicity allows them to diffuse through the plasma membrane (lipid bilayer) and bind their receptor in the cytoplasm or nucleus. Steroid hormones are labeled as first messengers because they perform the initial signaling that influences the nuclear transcription and cytoplasmic translation of physiologically required proteins.
- **Tyrosine derivatives** are derived from the amino acid tyrosine. These hormones function similarly to either steroid hormones (eg, thyroid hormones T3 and T4) or peptide hormones (eg, catecholamines epinephrine and norepinephrine), based on their structure.

(Choice A) The binding of a peptide hormone induces a signaling cascade by activating a *different* second messenger within the cell. Therefore, peptide hormones act as first (*not* second) messengers.

(Choices B and D) Steroid hormones, *not* peptide hormones, are able to dissociate from their carrier proteins in the blood and diffuse through the plasma membrane.

Educational objective:

Peptide hormones are amino acid-based, water-soluble molecules that travel freely through the bloodstream and act as first messengers by binding to an extracellular receptor on the target cell membrane, which leads to activation of intracellular second messengers.

Time spent: 0 sec
Subject:
Biology

QID: 400662
System:
Endocrine and Nervous Systems

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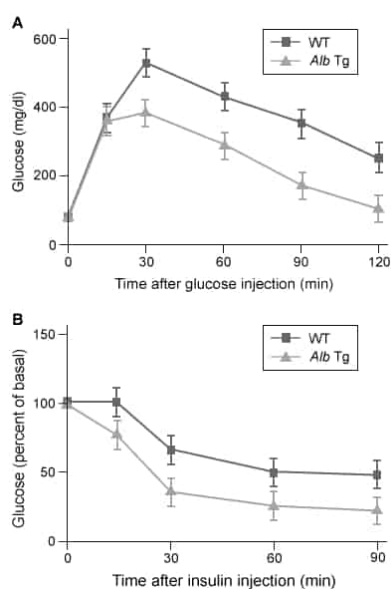


Figure 1 Effect of glucose and insulin administration on blood glucose of WT and *Alb* Tg mice fed a HFD. (Note: WT and *Alb* Tg mice fed a HFD had equal weight gain.)

Experiment 2

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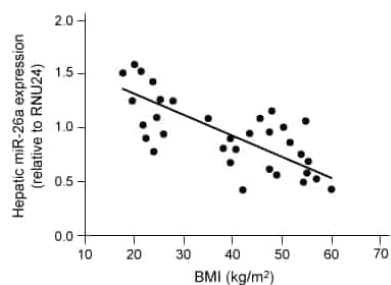
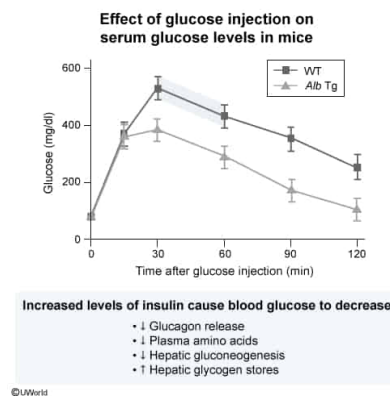


Figure 2 Relationship between BMI and hepatic miR-26a expression

During the interval between 30 and 60 minutes in Figure 1A, which of the following would be observed in WT mice?

- ☐ A. Increased glucagon release. [11%]
- ☐ B. Increased plasma amino acids. [2%]
- ☒ C. Increased hepatic glycogen stores. [79%]
- ☐ D. Increased hepatic gluconeogenesis. [7%]

Explanation:



Blood glucose levels are normally maintained within a narrow range by the **opposing effects** of insulin and glucagon.

- **High blood glucose** stimulates **insulin release**. Insulin **decreases blood glucose** concentration by inhibiting glucagon release, promoting glucose uptake by insulin-responsive tissues (ie, adipose, muscle), and decreasing hepatic glucose production (eg, **gluconeogenesis**). Insulin also **stimulates glycogen synthesis** (liver, muscle), triglyceride synthesis (via glucose transport into adipocytes), and protein synthesis (via entry of plasma amino acids into tissues).
- **Low blood glucose** stimulates **glucagon release**. Glucagon **increases blood glucose** levels by promoting production of gluconeogenic substrates and by stimulating glycogen breakdown (**glycogenolysis**) and hepatic gluconeogenesis. Glucagon limits glucose use by insulin-sensitive tissues and decreases glucose uptake by peripheral tissues. Low blood glucose has also been shown to promote the release of epinephrine, a hormone that stimulates glucagon release.

Figure 1A shows that blood glucose levels decreased between 30 and 60 minutes after the glucose injection in WT (normal) mice fed a HFD. This occurred because the glucose injection raised blood glucose levels and increased the release of insulin to lower blood glucose. **Insulin** decreased blood glucose in the **30–60-minute interval** by promoting **glucose uptake** in tissues throughout the body **by the liver** for glycogen synthesis, which **increased hepatic glycogen stores**.

The release of insulin would also *decrease* glucagon release (**Choice A**), *decrease* plasma amino acids by stimulating protein synthesis (**Choice B**), and *decrease* hepatic gluconeogenesis (**Choice D**).

Educational objective:

In the setting of high blood glucose, insulin is released in response and functions to decrease blood glucose levels. In contrast, glucagon is released in response to low blood glucose and functions to increase glucose levels.

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Biology

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Endocrine and Nervous Systems

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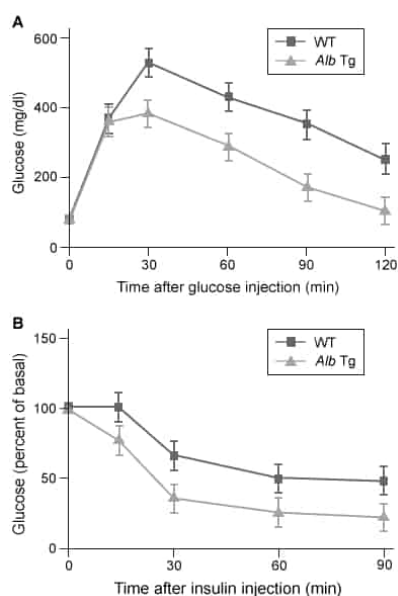


Figure 1 Effect of glucose and insulin administration on blood glucose of WT and *A/b* Tg mice fed a HFD. (Note: WT and *A/b* Tg mice fed a HFD had equal weight gain.)

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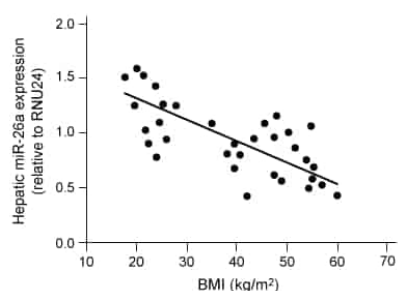


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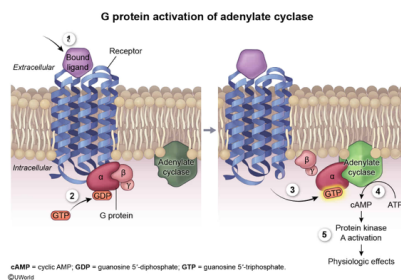
Recently fed *ob/ob* mice show deficient binding of glucagon to its G_s protein–coupled receptor, which most likely leads to:

- ☐ A. decreased cAMP to ATP conversion. [25%]
- ☐ B. upregulated adenylate cyclase activity. [6%]
- ☐ C. increased exchange of GTP for GDP. [8%]
- ☒ D. reduced protein kinase A activity. [59%]

Correct

59% answered correctly

Explanation:



Glucagon acts on target cells by binding its stimulatory **G protein–coupled receptor** on the cell membrane and inducing the **adenylate cyclase/cAMP second messenger cascade**. The G protein "attached" to the transmembrane cell surface receptor is composed of three subunits: alpha, beta, and gamma. When inactive, the alpha subunit (G_α) is bound to GDP, which is then replaced with GTP on ligand-receptor binding. The GTP-bound G_α dissociates from the beta and gamma subunits and proceeds to activate adenylate cyclase, an enzyme that catalyzes the conversion of ATP to cAMP, a second messenger. cAMP then activates protein kinase A, which subsequently phosphorylates the proteins necessary to produce ligand-specific physiological effects.

In the given scenario, the **downstream effects of deficient glucagon binding** in recently fed *ob/ob* mice would **decrease** the following: the exchange of GTP for GDP; adenylate cyclase activity; the conversion of ATP to cAMP; and **protein kinase A activity**.

(Choices A and B) Deficient binding of glucagon to its G protein–coupled receptor would decrease (*not* upregulate) adenylate cyclase activity, which would result in reduced conversion of ATP to cAMP (*not* cAMP to ATP).

(Choice C) GDP deactivates the G_α subunit, and GTP activates the G_α subunit. Activation is self-limited by hydrolysis of bound GTP to GDP by the G_α subunit. Decreased glucagon binding to its G protein–coupled receptor would lead to *decreased* activation of the G_α subunit, so a decreased (*not* increased) exchange of GTP (activating) for GDP (deactivating) would be observed.

Educational objective:

In the G protein–regulated cAMP signaling pathway, a ligand binds the transmembrane G protein–coupled receptor and activates the GDP-bound alpha subunit of the G protein by replacing GDP with GTP. The activated G alpha subunit activates the enzyme adenylate cyclase, which catalyzes the conversion of ATP into cAMP. Elevated cAMP leads to the activation of protein kinase A and subsequent signaling effects.

Time spent: 0 sec

Subject:
Biology

QID: 400664

System:
Endocrine and Nervous Systems

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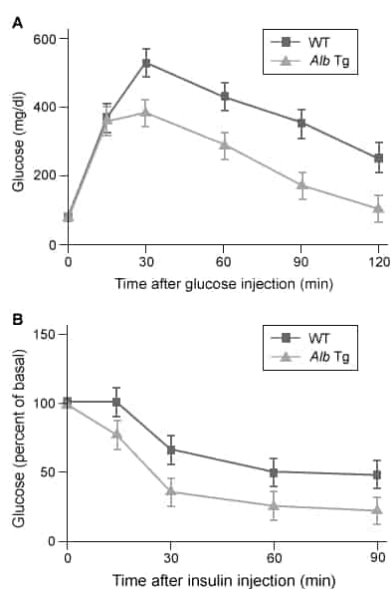


Figure 1 Effect of glucose and insulin administration on blood glucose of WT and *Alb* Tg mice fed a HFD. (Note: WT and *Alb* Tg mice fed a HFD had equal weight gain.)

Experiment 2

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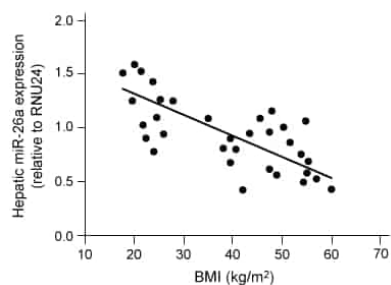


Figure 2 Relationship between BMI and hepatic miR-26a expression

The researchers hypothesize that miR-26a overexpression can rescue obesity-induced insulin resistance. Which experimental result in a follow-up study would NOT support their hypothesis?

- ☐ A. *Alb* Tg mice fed a HFD had the same levels of random and fasting insulin as WT mice fed a SHD. [12%]
- ☐ B. *Alb* Tg mice fed a HFD had the same levels of random and fasting glucose as WT mice fed a SHD. [20%]
- ☒ C. WT mice fed a HFD had higher insulin sensitivity than *Alb* Tg mice fed a HFD. [55%]
- ☐ D. WT mice fed a HFD had lower glucose tolerance than *Alb* Tg mice fed a HFD. [11%]

Correct

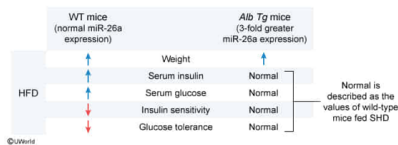
54% answered correctly

Explanation:

Insulin sensitivity refers to the biological response elicited by a fixed quantity of insulin. An organism is considered **insulin-sensitive** if only a minimal amount of insulin is needed to induce an appropriate reduction in glucose levels. In contrast, **insulin-resistant** organisms need substantially more insulin to take up the same amount of glucose.

In the experiment, mice fed a SHD are controls with normal weight and metabolism. However, mice fed a HFD over 14 weeks would likely exhibit obesity and obesity-induced insulin resistance (ie, decreased insulin sensitivity). This condition would be characterized by a *decrease* in glucose tolerance and an *increase* in levels of random/fasting insulin and glucose.

In this question, the researchers hypothesized that miR-26a overexpression (three-fold greater than wild-type [WT] in *A/b* Tg mice) can **rescue** obesity-induced insulin resistance. This hypothesis would be *supported* by the following findings in a follow-up study:



- If *A/b* Tg mice fed a HFD had similar levels of serum insulin and glucose as WT mice fed a SHD, this would mean that three-fold overexpression of miR-26a in *A/b* Tg mice was able to *decrease* insulin and glucose levels to the same levels as the control mice on a normal diet. As a result, *A/b* Tg mice fed a HFD would show improved (ie, decreased) insulin resistance as evidenced by a higher insulin sensitivity (**Choices A and B**).
- Glucose tolerance refers to the ability to avoid prolonged excessive glucose concentrations by rapidly clearing a large dose of glucose from the blood. Thus, if WT mice fed a HFD had lower glucose tolerance (clearance) than *A/b* Tg mice fed a HFD, this would support the claim that miR-26a overexpression is protective against insulin resistance (**Choice D**).

However, if **WT mice fed a HFD** were to show significantly **higher insulin sensitivity** than ***A/b* Tg mice fed a HFD**, this result would **fail to support the hypothesis** (and fail to replicate the results in Experiment 1) because it implies that miR-26a overexpression in *A/b* Tg mice does *not* rescue obesity-induced insulin resistance.

Educational objective:

An organism is considered insulin sensitive if only a minimal amount of insulin is needed to induce an appropriate reduction in glucose levels. In contrast, insulin-resistant organisms need substantially more insulin to take up the same amount of glucose.

Time spent: 0 sec
Subject:
Biology

QID: 400665
System:
Endocrine and Nervous Systems

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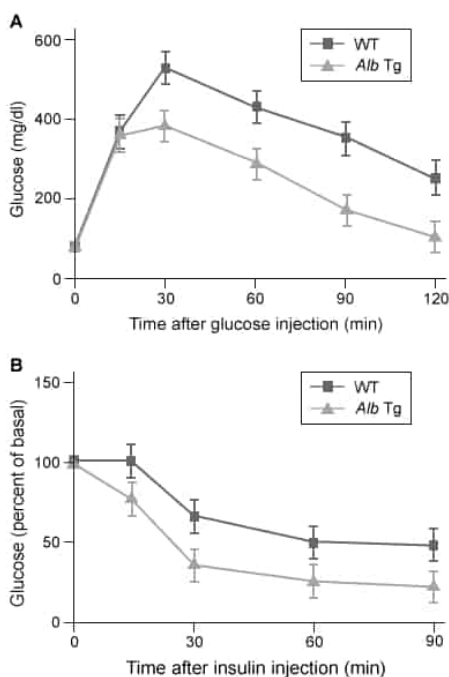


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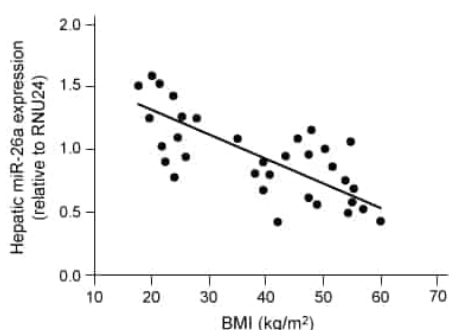


Figure 2 Relationship between BMI and hepatic miR-26a expression

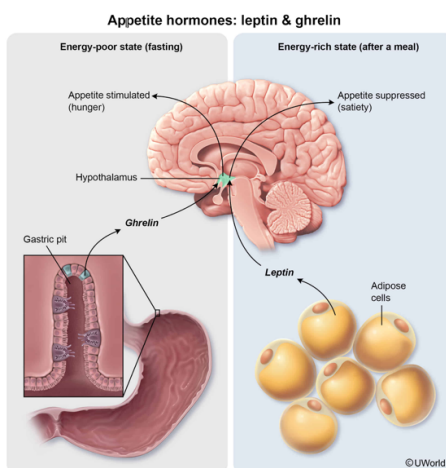
One of the human study subjects with a BMI of 56 kg/m^2 (Figure 2) is discovered to have a pathology similar to *ob/ob* mice. Like *ob/ob* mice, individuals affected by this pathology exhibit normal weight at birth but become severely obese afterward and have undetectable levels of leptin in the serum. What is the best explanation for the absence of serum leptin in this subject?

- ☐ A. Leptin cannot bind its receptor in the hypothalamus. [13%]
- ☐ B. Leptin cannot bind its receptor in the thyroid gland. [8%]
- ☐ C. Leptin cannot be secreted from gastric cells. [24%]
- ✓ ☒ D. Leptin cannot be secreted from adipocytes. [53%]

Correct

53% answered correctly

Explanation:



After a meal, the body is in an **energy-rich state** (ie, high concentrations of glucose and lipids); as a result, the hormone **leptin** is released by white **adipose tissue**. This reaction triggers feelings of **satiety** by

communicating to the hypothalamus that the stomach is full, thereby **suppressing appetite**. In general, the greater the adipose tissue stores, the higher the leptin levels in the serum.

The passage states that *ob/ob* mice are leptin deficient, signifying that white adipocytes are unable to produce/release leptin, which in turn causes undetectable serum leptin levels. Consequently, the hypothalamus never receives the satiety signal (ie, leptin) to suppress appetite and cease food consumption, leading to severe obesity. An obese individual (eg, BMI of 56 kg/m^2) with a pathology similar to *ob/ob* mice would therefore not likely achieve adequate appetite suppression due to **absent leptin production**. This person would experience constant hunger and engage in food-seeking behavior and food consumption, which would in turn cause continuous weight gain.

(Choices A and B) In leptin deficiency, whether or not leptin can bind its receptor would *not* be the reason why leptin levels are undetectable in the serum. Instead, the absence of leptin in the circulation would be caused by a lack of leptin *production* by white adipocytes.

(Choice C) In a fasting or energy-poor state (ie, low concentrations of glucose and lipids), the hormone ghrelin (*not* leptin) is released by gastric cells in the stomach wall. This ghrelin release triggers feelings of hunger by communicating to the hypothalamus that the stomach is empty, thereby increasing appetite.

Educational objective:

In an energy-rich state (eg, after a meal), leptin is released by white adipocytes to trigger appetite suppression via the hypothalamus. In contrast, in an energy-poor state, ghrelin is released by stomach gastric cells to trigger hunger and food-seeking behavior via the hypothalamus.

Time spent:0 sec
Subject:
Biology

QID:400666
System:
Endocrine and Nervous Systems